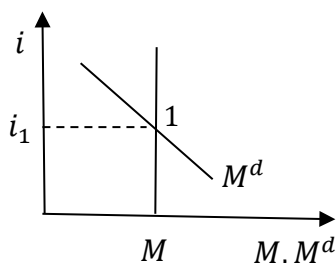


23 giugno 2020_matt – Sketch of answers

Question 1 (5 points)

In country Alpha, the central bank chooses, and wants to keep constant at the chosen level, the interest rate, while in country Beta the central bank chooses the money supply M (and, once again, keeps it constant at the level it has chosen). The figure below depicts the money market equilibrium in the two countries. As you can see, in this initial equilibrium position money supply is the same in both, and the same is true about the money demand curve and the equilibrium interest rate. Explain if you agree, or do not agree, with the following statement: “If, in both countries, income Y rises by the same amount, the central bank of Alpha will have to increase the money supply to keep the interest rate constant, while in Beta – whose central bank wants to prevent the money supply from changing – the interest rate will fall”.

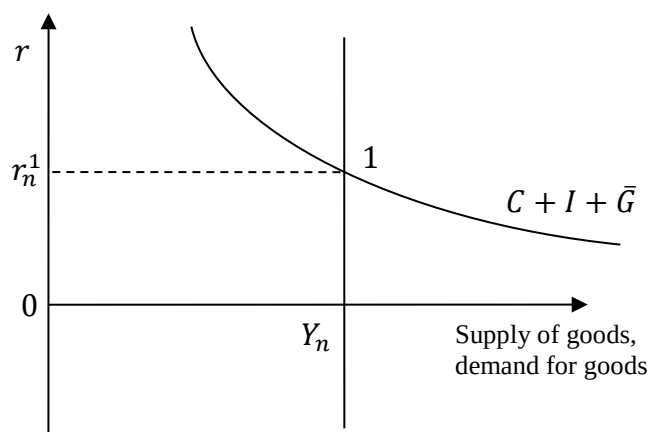


SKETCH OF ANSWER

False. The rise in income leads individuals to demand more money for any given level of the interest rate, so that the M^d curve shifts to the right. While it is true that, in Alpha, to keep the interest rate constant, the central bank will have to increase money supply by the same amount by which money demand has risen, in Beta – whose central bank keeps the money supply constant – the interest rate will go up, not down.

Question 2 (5 points)

By making reference to the graph “supply of goods-demand for goods” used to analyse the determination of the medium-run equilibrium, natural real interest rate, r_n , and reproduced below for your convenience, explain if and how r_n will change following an increase in firms’ market power that leads to a higher *mark-up* (a higher value for the parameter m in the WS-PS model of your textbook). In addition, compare the composition of aggregate demand prevailing after this change with that prevailing before the increase in the *mark-up*. Explain.



SKETCH OF ANSWER

In the WS-PS model, an increase in the mark-up shifts the PS curve downwards and raises the natural rate of unemployment, thus leading to a decrease in the associated natural level of production, Y_n . In the graph above, the supply curve would shift to the left, going through this new, smaller Y_n , while the demand curve would remain unchanged. It follows that, in the new medium run equilibrium, the real interest rate will go up by the amount needed to make sure that the demand for goods falls exactly by the same amount by which the supply of goods has gone down. Finally, in this new equilibrium consumption will be lower (since income is lower), investment lower (due to the rise in the real interest rate and the fall in Y_n) and G , being exogenous, unchanged.

Question 3 (5 points)

Explain if, and why, you agree, or do not agree, with the following statement [when answering, assume that, in each time period, the economy is described by a static IS-LM model, with consumption and investment depending on contemporaneous variables only, and that **the central bank chooses the money supply**]:

“The prices of stocks at time t , Q_t , will unambiguously rise if, from time t onwards, government purchases of goods and services, G , go up”.

SKETCH OF ANSWER

The statement is incorrect. If the central bank chooses the money supply, so that the LM curve is positively sloped, a permanent fiscal expansion that shifts the IS curve to the right leads in every period to a new equilibrium where both income and the interest rate have gone up. Higher income levels lead to higher expected dividends, but higher interest rates lower the present value of any given stream of future expected dividends. It follows that the effect of the permanent increase in G on the prices of stocks at time t will be uncertain.

Question 4 (5 points)

In country Macrolandia there are only one-year and two-year bonds. The current and the expected future inflation rates are both zero, so that the real and the nominal interest rates are equal. In period $t + 1$ (“the future”), a standard IS-LM model describes the functioning of the economy. In particular, in the year $t + 1$ consumption depends only on disposable income at $t + 1$, and investment on the interest rate and output at $t + 1$ only. Turning now to the current period (t , “the present”), consumption is increasing in both current and expected future disposable income, $C_t = C(Y_t - \bar{T}_t, Y_{t+1}^e - \bar{T}_{t+1}^e)$, investment depends positively on current and expected future income levels, and negatively on the current and expected future levels of short-term interest rates, the demand for money has the usual functional form, and G and T are exogenous as usual. Starting from an initial equilibrium, at t the central bank of Macrolandia, that chooses the interest rate, announces that at time t , and at time t only, will raise the policy rate by two percentage points. If in Macrolandia, at time t the yield curve is initially horizontal, and assuming that the announcement is unexpected and believed by the individuals in the economy, which of the following three figures (where the new curve is labelled “after”), describes the effect of the announcement on Macrolandia’s yield curve? Explain in detail, describing the reasons for any change in each of the two points that the yield curve connects. At time t , will the yield to maturity on two-year bonds change? If so, by how much? If not, why?

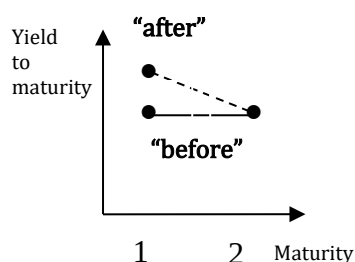


FIGURE 1

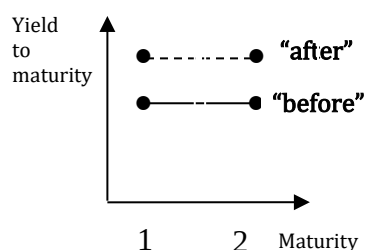


FIGURE 2

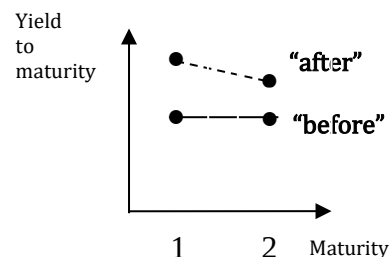


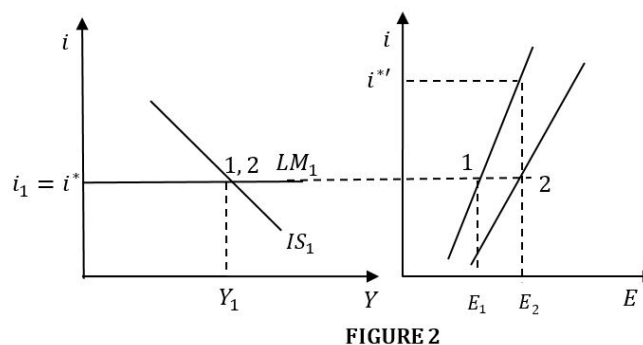
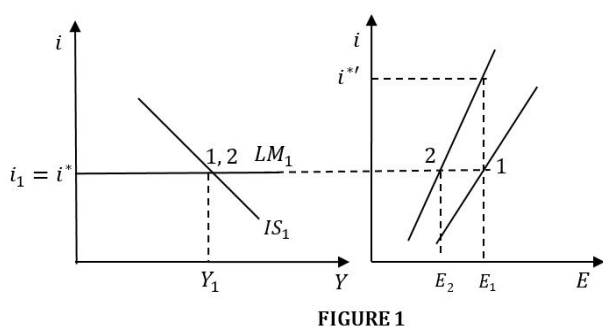
FIGURE 3

SKETCH OF ANSWER

The correct figure is Figure 3. The central bank is raising i_{1t} by 2%, while keeping the future short rate unchanged. Of the two points that the yield curve connects, the first one will therefore go up by two percentage points, while the second one, i_{2t} , by one percentage point. In fact, the current yield on two-year bonds is equal to the average of the current and the future expected short term interest rates, and $i_{1,t+1}^e$ does not change. The new yield curve will lie above the old one and will have a negative slope, signaling that short rates will be on average higher than before, and that their increase takes place in the current period.

Question 5 (6 points)

Consider an open economy under a flexible exchange rate regime, described by a standard ‘open economy IS-LM’-‘interest parity condition’ model. Domestic and foreign prices are constant - for simplicity, both equal to 1 ($P = P^* = 1$) - and the country’s central bank chooses the interest rate, initially setting it at the same level as the foreign one, so that in the initial equilibrium (point ‘1’ in the figures below) $i_1 = i^*$, $Y = Y_1$ and $E = E_1$. Taking these assumptions into account, and having explained which value the expected future exchange rate must necessarily take on in this economy, suppose that the foreign interest rate, i^* , goes up (in the figures, it becomes $i^{*'} > i^*$), while the domestic interest rate chosen by the central bank, foreign output Y^* and all the remaining exogenous variables - E^e included - remain constant. Assuming that the country’s government decides to change taxes to prevent domestic output Y from varying, which of the two graphs below correctly depicts the new equilibrium (point ‘2’ in the figures) that will be reached following the exogenous increase in i^* and the associated fiscal policy intervention just described? To achieve its aim, should the government raise or cut taxes? Finally, carefully explain how the exchange rate, the money supply, consumption, investment and net exports will change as the economy moves from the initial equilibrium position (‘1’) to the final one (point ‘2’).



SKETCH OF ANSWER

From the interest parity condition it follows that, if $i = i^*$, then $E^e = E_1$, where E_1 is the value that the exchange rate takes on in the initial equilibrium. In the right-hand panel of the graph, the decrease in the foreign interest rate leads to a leftward shift in the line representing the interest parity condition; the new line will now go through – as it has to do by construction – the point ($E = E^e = E_1, i = i^{*'} \text{)$, where $i^{*'}$ is the new level of the foreign interest rate. It follows that the correct figure is Figure 1. The ensuing depreciation of the exchange rate, that now becomes E_2 , would raise net exports, thus shifting the IS to the right and increasing equilibrium output. To prevent Y from changing, the government will have to raise taxes by the amount needed to keep the IS in the initial position. As mentioned before, in the new equilibrium (point 2 in the two panels) the exchange rate will be lower, since the increase in the foreign interest rate leads to a higher demand for foreign assets, and therefore for the currency in which they are denominated; consumption will be lower (Y is unchanged, but taxes are higher); investment is unchanged, just like Y and the domestic interest rate; with Y and i unchanged, both money demand and money supply will be unchanged; finally, net exports will be higher, due to the exchange rate depreciation and the constancy of Y . [Even though the question does not ask for this extra step, it is immediate to reach the conclusion that the above discussion implies that, in the new equilibrium, net exports will have necessarily gone up by the same amount by which consumption has gone down].

Question 6 (5 points)

Suppose now that, in the economy studied in the previous question, investment is entirely exogenous, rather than a function of i and Y , as in the standard case just considered. How will the conclusion you have reached about the effects of the increase in the foreign interest rate and the simultaneous fiscal policy intervention aimed at keeping Y constant change in this case? Explain.

SKETCH OF ANSWER

The effects will be identical to those described when answering the previous question. The exchange rate depreciation will continue to stimulate the country's net exports, thus shifting to the right an IS curve that will still be negatively sloped, due to the fact that E depends – via interest parity condition – on the domestic interest rate. To prevent Y from changing, the government will have to bring back the IS to the left by raising net taxes. In the new equilibrium, and as before, C will be smaller, NX larger, G unchanged and, of course, I unchanged.